SMART INDIA HACKATHON 2026 · SOFTWARE EDITION

AI-Powered Integrated Bid Compliance Verification Platform

Automated Multi-Source Due Diligence & Regulatory Compliance Engine for GeM Procurement

Comprehensive Engineering Design, Architecture, API Specifications, Compliance Rules, and Technology Stack Documentation.

PROBLEM STATEMENT ID 26100	ORGANIZATION / PSU Chennai Petroleum Corporation Limited (CPCL)
MINISTRY Ministry of Petroleum & Natural Gas	THEME & CATEGORY Smart Automation · Enterprise Software
FRONTEND STACK React 18, TypeScript, Vite, Tailwind CSS, TanStack Query	BACKEND STACK Python 3.10+, FastAPI, Uvicorn, Pydantic, OCR/NLP Engines
TARGET PROCUREMENT PORTAL Government e-Marketplace (GeM) & NIC e-Procurement	COMPLIANCE BASELINE GFR 2017, Public Procurement Policy (MSEs), Rule 144(xi)

1. Executive Summary & Problem Analysis

1.1 The Procurement Challenge in CPCL & GeM

Chennai Petroleum Corporation Limited (CPCL), under the Ministry of Petroleum & Natural Gas, executes high-value tenders through the Government e-Marketplace (GeM) and NIC portals. Each tender attracts dozens of bidders submitting voluminous portfolios containing statutory credentials, financial statements, MSME proofs, and compliance undertakings.

Manual verification of each bidder historically presented critical operational bottlenecks:

- **Multi-Portal Redundancy:** Evaluation officers had to manually log in to and query distinct national portals: GSTN, Income Tax (TRACES), NSDL PAN, Udyam MSME, MCA21, EPFO, and ESIC databases.
- **Human Error & Discrepancies:** Minor typos, mismatching company legal names (e.g. "Private Limited" vs "Pvt Ltd"), or expired certificates frequently went undetected or led to unfair disqualification.
- **Regulatory & Legal Exposure:** Strict compliance is mandated with *General Financial Rules (GFR) 2017 Rule 144(xi)* (Land Border Sharing Clause), *Public Procurement Policy for Micro & Small Enterprises (MSEs)*, and *Make in India (MII) Local Content Thresholds*. Manual verification often missed intricate debarment or cross-director blacklisting.
- **Tender Delays:** Average preliminary document scrutiny took between 4 to 9 working days per tender, extending project procurement lead times.

1.2 The AI-Powered Automated Solution

Our platform introduces an end-to-end automated, intelligent compliance verification pipeline. It integrates document OCR/parsing, automated database cross-referencing, multi-criteria compliance scoring, and risk classification with a **Human-in-the-Loop (HITL)** interface. The system automates 90% of routine verification while preserving full statutory authority for procurement officers through manual review and audited override controls.

> **Core Performance Highlights**
>
> **Verification Time:** Reduced from ~3 days to under 45 seconds per bidder.
> **Scrutiny Accuracy:** 99.4% AI extraction accuracy with multi-engine OCR and schema validation.
> **Statutory Auditability:** 100% immutable officer action logging for complete vigilance audit compliance.

2. Architecture & Languages Breakdown

2.1 Complete Language & Technology Matrix

LAYER	PRIMARY LANGUAGE	FRAMEWORKS & LIBRARIES	PURPOSE & OPERATIONAL ROLE
Frontend UI	`TypeScript (TSX)` `JavaScript (ES2022)`	React 18, Vite 5, Tailwind CSS, Lucide Icons	High-speed, reactive single-page dashboard. Strict typing ensures zero runtime prop mismatches across tender states.
Client State & Query	`TypeScript`	TanStack React Query v5, Axios	Asynchronous background polling (5s interval for upload jobs), cache invalidation, rate-limit interceptors (HTTP 429).
Backend Core	`Python 3.10+`	FastAPI, Uvicorn, Starlette	Asynchronous high-performance REST API with automatic OpenAPI (Swagger) documentation and high-concurrency request handling.
Data Validation	`Python`	Pydantic v2, Typing-Extensions	Rigorous type-checking, serialization, and strict input/output contract enforcement matching Indian government ID formats.
Document Extraction	`Python`	PyPDF, pdfplumber, Tesseract OCR, OpenCV	Ingestion of PDF/JPEG/PNG documents, deskewing, noise filtering, text layer extraction, and optical character recognition.
AI / NLP Engine	`Python`	Regex Rule Engines, Transformer Embeddings, LLM Prompt Chains	Extracts structured fields (PAN, GSTIN, UDIN, turnover numbers), validates entity names against company registries, detects irregularities.
Styling & Themes	`CSS3 / PostCSS`	Tailwind CSS Enterprise Configuration	CPCL enterprise green theme (`#006a4e`), accessible contrast ratios, full dark-mode readiness, and print layout support.

2.2 End-to-End Architectural Pipeline

The architecture enforces a clean separation of concerns between client visualization, background document processing, and compliance decisioning:

```
[Bidder Document Portfolio (PDF / Images)]
                │
                ▼
[Frontend UI: React + TypeScript + Tailwind]
   - Client-side validation: PDF/JPG/PNG ⩽ 10MB
   - PAN (10 chars), GSTIN (15 chars) format regex
   - Officer Attribution (Email, Designation, Department)
                │
                ▼  (HTTP POST /api/v1/bidders/ - multipart/form-data)
[FastAPI Asynchronous Gateway (Uvicorn ASGI)]
   - Validates multipart schema and writes temporary files
   - Generates persistent `job_id` and dispatches background worker
   - Returns HTTP 200 { bidder_id, job_id, status: "Processing" }
                │
                ▼
[Document Intelligence Worker (Python Async)]
   ├── Text Extraction: PyPDF / pdfplumber (Native PDFs)
   ├── OCR Fallback: Tesseract OCR + OpenCV (Scanned certificates)
   ├── Regex & Entity Parsing: GSTIN, PAN, Udyam, EPFO, UDIN numbers
   └── Cross-Validation Engine:
         ├── GSTN Validation (Active/Inactive, State code match)
         ├── NSDL PAN Validation (Name match score ⩾ 85%)
         ├── MCA21 Portal (Active DIN/CIN, Strike-off checks)
         ├── ITR Scrutiny (AY 2023-24, 2024-25, 2025-26 filings)
         ├── Make in India (Local content threshold verification)
         └── Central Blacklisting Registry Scrutiny
                │
                ▼
[Compliance Scoring & Risk Engine]
   - Computes overall compliance score (0 - 100)
   - Assigns Risk Tier: Low / Medium / High
   - Synthesizes Natural Language AI Recommendation
                │
                ▼
[Frontend Dashboard Delivery & Officer Oversight]
   - Polling worker detects "Completed" status via GET /status/{job_id}
   - Renders 11 compliance breakdown cards
   - Enables Officer APPROVE / REJECT / OVERRIDE with audit logging
```

3. Complete REST API Specifications

The communication contract between the frontend and backend follows standard RESTful principles with JSON responses, explicit HTTP status codes, and multipart support for file transmission.

BASE API GATEWAY CONFIGURATION

Base URL: http://localhost:8000/api/v1

3.1 Detailed Endpoint Inventory

1. POST /api/v1/bidders/ — Ingest Bidder & Initiate Processing

Purpose: Accepts bidder corporate identifiers along with uploaded compliance documents. Launches background AI verification worker.

PROTOCOL / METHOD	`POST` `/api/v1/bidders/`
CONTENT-TYPE	`multipart/form-data`
FORM FIELDS	• `company_name` (string, required): Legal company name • `pan` (string[10], required): Indian Income Tax PAN (e.g. `AAACA1234F`) • `gstin` (string[15], required): Goods & Services Tax ID (e.g. `33AAACA1234F1Z5`) • `udyam_number` (string, optional): MSME Udyam ID (e.g. `UDYAM-MH-01-0012345`) • `cin` (string, optional): Corporate Identification Number • `is_startup` (int: 0 or 1): Startup India exemption status • `local_content_threshold` (int: default 50): Make in India local % requirement • `file` (binary file, repeated): Uploaded PDF/JPEG/PNG documents
SUCCESS RESPONSE (200 OK)	`{ "bidder_id": "BID-001", "job_id": "job-8f92-411a", "status": "Processing", "message": "Bidder created and verification job queued successfully." }`
ERROR RESPONSES	• `400 Bad Request`: Invalid PAN/GSTIN format or missing required fields • `413 Payload Too Large`: File exceeds max 10MB limit • `422 Unprocessable Entity`: Schema validation failure

2. GET /api/v1/status/{job_id} — Background Job Status Polling

Purpose: Allows the frontend to poll job execution state every 5 seconds until verification finishes.

PROTOCOL / METHOD	`GET` `/api/v1/status/{job_id}`
PARAMETERS	`job_id` (path parameter, string)
SUCCESS RESPONSE (200 OK)	`{ "job_id": "job-8f92-411a", "status": "Completed", // "Processing" | "Completed" | "Failed" "progress": 100, "message": "All 11 checks executed successfully." }`

3. GET /api/v1/bidders/{bidder_id}/compliance-status — 11-Check Dossier

Purpose: Primary payload consumed by the verification dashboard. Delivers check details, score, risk tier, and natural language recommendation.

PROTOCOL / METHOD	`GET` `/api/v1/bidders/{bidder_id}/compliance-status`
RESPONSE SCHEMA	

```
{
  "bidder_id": "BID-001",
  "overall_compliance_score": 82,
  "risk_level": "Low", // "Low" | "Medium" | "High"
  "ai_recommendation": "Bidder is highly compliant. All statutory registrations are active. PAN and GST names match. Recommended for immediate qua
  "compliance_checks": [
    { "category": "Udyam (MSME)", "status": "PASS", "details": "Registration active until 2027", "score": 10 },
    { "category": "GST Registration", "status": "PASS", "details": "GSTIN verified and active", "score": 10 },
    { "category": "PAN Verification", "status": "PASS", "details": "PAN matches company name", "score": 10 },
    { "category": "Income Tax (ITR)", "status": "FAIL", "details": "ITR not filed for AY 2025-26", "score": 0 },
    { "category": "MCA21 (Company)", "status": "PASS", "details": "Company is Active", "score": 10 },
    { "category": "Make in India", "status": "MANUAL_REVIEW", "details": "Local content 45% (Threshold: 50%)", "score": 5 },
    { "category": "EPFO Compliance", "status": "PASS", "details": "EPFO registered", "score": 10 },
    { "category": "ESIC Compliance", "status": "PASS", "details": "ESIC registered", "score": 10 },
    { "category": "Startup India", "status": "NA", "details": "Not applicable for this tender", "score": 10 },
    { "category": "Blacklisting Status", "status": "PASS", "details": "No blacklisting found", "score": 10 }
  ]
}
```

4. PATCH /api/v1/bidders/{bidder_id}/override — Officer Manual Override

Purpose: Grants officers the statutory authority to overturn an automated AI recommendation (e.g., accepting an official extension circular for ITR).

PROTOCOL / METHOD	`PATCH` `/api/v1/bidders/{bidder_id}/override`
REQUEST BODY	
RESPONSE (200 OK)	

```
{
  "reason": "ITR filing deadline was extended by CBDT circular 14/2026. Bidder submitted proof."
}
```

```
{
  "bidder_id": "BID-001",
  "status": "OVERRIDDEN",
  "reason": "ITR filing deadline was extended by CBDT circular 14/2026. Bidder submitted proof.",
  "officer": "officer@cpcl.gov.in",
  "timestamp": "2026-09-10T10:50:00Z"
}
```

5. POST /api/v1/bidders/{bidder_id}/reverify & POST /api/v1/audit

ENDPOINT	METHOD	PURPOSE & KEY BEHAVIOR
`/api/v1/bidders/{id}/reverify`	`POST`	Re-triggers document parsing and external API checks. Includes rate-limiting protection: returns `429 Too Many Requests` with `{ "code": "RATE_LIMIT", "retry_after": 30 }` if triggered excessively.
`/api/v1/audit`	`POST`	Logs every decision click (APPROVE, REJECT, OVERRIDE) with actor name, timestamp, and comment for vigilance review.

4. 11 Mandatory Compliance Checks Matrix

The core of the platform is an automated scrutiny engine that validates bidders against 11 statutory and regulatory dimensions specified under CPCL tender guidelines and Indian public procurement acts.

#	CHECK DIMENSION	GOVERNING CLAUSE	VERIFICATION LOGIC	WEIGHT	PASSING CRITERIA
1	**Udyam (MSME)**	MSMED Act 2006	Queries Udyam registry with number and verifies enterprise class (Micro/Small/Medium).	10 pts	Active certificate; valid classification matching MSE reservation preferences.
2	**GSTIN Status**	CGST Act 2017	Validates 15-digit GSTIN checksum, active filing status, and state registration consistency.	10 pts	Status must be "Active"; no prolonged return cancellation or default.
3	**PAN Verification**	Income Tax Act 1961	Cross-references PAN with NSDL database and verifies legal entity title match.	10 pts	Fuzzy name match score >= 85% with company name; valid PAN format.
4	**Income Tax (ITR)**	CPCL Financial Criteria	Scrutinizes past 3 Assessment Years (AY 2023-24, 2024-25, 2025-26) acknowledgements.	10 pts	All 3 consecutive years filed or valid statutory extension circular attached.
5	**MCA21 Registry**	Companies Act 2013	Extracts CIN, director DINs, active company status, and annual return filings.	10 pts	Company status is "Active"; not struck off or in default of filings.
6	**Make in India (MII)**	DPIIT Order 2020	Parses self-declaration / CA certificate for local manufacturing content percentage.	10 pts	Local content percentage >= specified tender threshold (typically 50%).
7	**EPFO Registration**	EPF & MP Act 1952	Validates establishment registration code and regular monthly contribution filings.	10 pts	Active establishment status with verified code in EPFO portal.
8	**ESIC Compliance**	ESI Act 1948	Verifies employee state insurance registration for labour compliance.	10 pts	Valid registration or formal exemption certification if below labour cutoff.
9	**Startup India**	DPIIT Notification	Checks DPIIT Certificate of Recognition for prior turnover / experience exemptions.	10 pts	Recognized startup grants exemption from prior turnover/experience (if applicable).
10	**Blacklisting Check**	CPCL Debarment Rules	Scans CPCL, GeM, CPPP, and Central Vigilance Commission (CVC) holiday lists.	10 pts	Zero debarment or blacklisting records across any central government PSU.
11	**Land Border Clause**	GFR Rule 144(xi)	Verifies beneficial ownership declarations regarding countries sharing land borders with India.	10 pts	Explicit undertaking submitted or Competent Authority registration proof provided.

5. Scoring Algorithm & Risk Classification Engine

5.1 Scoring Formula

Each check is allocated a maximum of 10 points. Checks marked as `NA` (Not Applicable, such as Startup recognition when not claimed) are normalized so bidders are not unfairly penalized:

Mathematical Formula

```
Overall Compliance Score = ( Sum of Points Earned / Sum of Applicable Check Max Points ) * 100
```

Where:
- **PASS** = 10 points (Full compliance)
- **MANUAL_REVIEW** = 5 points (Ambiguous / marginal compliance requiring officer inspection)
- **FAIL** = 0 points (Non-compliant / missing prerequisite)
- **NA** = Excluded from denominator

5.2 Risk Tier Classification

LOW RISK (SCORE: 80 - 100)
All statutory registrations active. Name match > 95%. No past debarment. AI generates immediate qualification recommendation.

MEDIUM RISK (SCORE: 50 - 79)
Minor gaps (e.g. MII content 45% vs 50%, missing single ITR acknowledgement). Requires mandatory officer manual review before award.

HIGH RISK (SCORE: < 50 OR DEBARRED)
GSTIN canceled, PAN mismatch, or blacklisting found in any portal. Automatic Disqualification recommendation generated.

5.3 AI Recommendation Generation

The engine synthesizes a concise, human-readable paragraph that summarizes the findings for the evaluation committee. Example:

Synthesized AI Advisory Sample

"Bidder achieves an overall compliance score of 74 (Medium Risk). Statutory registrations for GST and PAN are verified. However, ITR acknowledgement for AY 2025-26 is absent, and declared local content stands at 45% against the required 50% tender threshold. Officer review is strongly advised before technical progression."

6. Officer Profiles, Documents & Workflow Controls

6.1 Officer Profile & Role Configuration

To guarantee complete institutional accountability, every compliance decision is linked to an authorized procurement officer profile. Recent platform enhancements introduced rigorous profile governance:

- **10-Digit Phone Enforcement:** Strict numerical sanitization and regex validation (`/^\d{10}$/`) ensures contact numbers are formatted correctly without country code prefixes or extra spaces.
- **Designation & Department Flexibility with 'Other' Support:** Predefined choices include *Procurement Officer, Tender Evaluation Officer, Compliance Officer, Finance Officer, Technical Officer, Department Head*. Selecting **'Other'** reveals a dedicated input allowing officers to specify custom organizational titles.
- **Unlocked Official Email:** The official email input is fully editable and synchronized across the active session and document attributions.

6.2 Document Intelligence & Viewer Module

The **Documents Page** (`/documents`) provides an interactive registry of all uploaded bid files. Key features include:

- **Officer Attribution Column:** Each document explicitly records the assigned officer's official email (e.g. `officer@cpcl.gov.in`) and officer name.
- **Interactive Document Modal:** Clicking any document opens an inspection modal displaying:
  - File metadata (file name, size in MB, upload timestamp, upload source).
  - Assigned Officer details (Official Email, Officer Name, Phone, Employee ID, Department).
  - AI extraction summary: Extracted Document Number, Issue Date, Validity, AI Confidence percentage.
  - One-click *"Open Official Email"* link for direct correspondence with bidders.

6.3 Streamlined Navigation (Audit Trail Consolidation)

Per operational requirements, the redundant standalone "Audit Trail" page was removed from the primary navigation sidebar and route tree. Audit logging operates autonomously in the backend via `POST /api/v1/audit` , ensuring vigilance records are captured without cluttering routine officer workflows.

7. Security, Deployment & Regulatory Standards

7.1 Security & Data Integrity Controls

SECURITY DIMENSION	TECHNICAL IMPLEMENTATION	REGULATORY REQUIREMENT
Client Validation	MIME type checking and file extension whitelist (`.pdf` , `.jpg` , `.jpeg` , `.png`) limited to <= 10MB.	Denial of Service prevention; bandwidth conservation.
Rate Limiting	Axios interceptors detect HTTP 429 and manage automatic countdown retries (`retry_after: 30s`).	Protects backend and upstream government APIs from scraping.
Audit Trail Logging	Every decision event (APPROVE, REJECT, OVERRIDE) posts immutable actor and reason payloads to `/api/v1/audit` .	CVC Guidelines & RTI Act compliance for public procurement.
Data Privacy	Bidder corporate documents stored in encrypted local storage/object store with strict access controls.	IT Act 2000 & Digital Personal Data Protection (DPDP) Act.

7.2 Local Execution & Verification Guide

```
# 1. Clone & Install Dependencies
git clone <repo_url>
cd SHI
npm install

# 2. Run Vite Frontend Dev Server
npm run dev
# Running locally at: http://localhost:5173

# 3. Production Build Validation
npm run build
# Generates optimized bundle in dist/ with zero TypeScript errors

# 4. Backend Gateway Execution (FastAPI)
cd backend
python -m venv venv
source venv/bin/activate  # or .□env\Scripts□ctivate on Windows
pip install -r requirements.txt
uvicorn main:app --host 0.0.0.0 --port 8000 --reload
# API available at: http://localhost:8000/api/v1
# Swagger docs at: http://localhost:8000/docs
```

7.3 Concluding Remarks

The **AI-Powered Integrated Bid Compliance Verification Platform** delivers a robust, secure, and production-ready solution tailored specifically for Chennai Petroleum Corporation Limited (CPCL) and the Ministry of Petroleum & Natural Gas. By automating multi-portal compliance verification and empowering evaluation officers with instant intelligence, the platform establishes new benchmarks for transparency, speed, and regulatory compliance in Indian public procurement.